Research Proposal

Explainable Artificial Intelligence (XAI) for Sustainable Housing Policy: A Data-Driven Approach Using the USA Real Estate Dataset

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1. Introduction

The global housing crisis presents one of the most significant socioeconomic challenges of the 21st century. Rapid urbanization, coupled with rising property prices and environmental concerns, demands data-driven approaches to inform sustainable housing policies. According to the United Nations Sustainable Development Goal 11 (SDG 11), making cities and human settlements inclusive, safe, resilient, and sustainable requires transparent, evidence-based policymaking (United Nations, 2015).

Machine learning (ML) models have demonstrated strong predictive capabilities in real estate valuation (Park & Bae, 2015; Antipov & Pokryshevskaya, 2012). However, the "black-box" nature of complex ML models limits their adoption in public policy, where stakeholders require transparency and interpretability to build trust in model-driven recommendations (Arrieta et al., 2020).

Explainable Artificial Intelligence (XAI) addresses this challenge by providing methods to interpret and understand model predictions (Molnar, 2020). Techniques such as SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017) and LIME (Local Interpretable Model-agnostic Explanations) (Ribeiro et al., 2016) enable both global and local interpretability, making ML models suitable for informing public policy decisions.

This proposal outlines a research project that applies XAI techniques to housing price prediction using the USA Real Estate Dataset, with the goal of deriving actionable, transparent insights for sustainable housing policy.

2. Problem Statement

Despite the availability of large-scale real estate data, housing policy decisions remain largely based on traditional statistical methods and expert judgment. Key challenges include:

1. **Lack of transparency**: Advanced ML models that achieve high predictive accuracy are often opaque, making it difficult for policymakers to understand *why* certain properties are valued higher or lower.
2. **Limited sustainability metrics**: Existing real estate analyses rarely incorporate sustainability-relevant features such as land use efficiency, housing density, or environmental indicators.
3. **Regional inequities**: Housing markets vary significantly across states and regions, yet policies are often formulated without data-driven understanding of these disparities.
4. **Prediction fairness**: ML models may exhibit systematic biases across different price ranges or geographic regions, potentially reinforcing existing inequities.

This research aims to address these gaps by developing an XAI-based framework that provides transparent, interpretable, and policy-relevant insights from large-scale housing data.

3. Research Objectives

The primary objectives of this research are:

1. **Develop predictive models** for housing prices using multiple machine learning algorithms, including baseline models (Linear Regression, Decision Tree) and advanced ensemble methods (Random Forest, XGBoost).
2. **Apply XAI techniques** (SHAP and LIME) to interpret model predictions and identify the key factors driving housing prices at both global and local levels.
3. **Engineer sustainability-relevant features** such as land use efficiency, room density, and affordability indices to enrich the analysis with domain-specific metrics.
4. **Conduct rigorous model evaluation** using 5-fold cross-validation, ablation studies (to quantify the impact of engineered features), and fairness analysis across price ranges and geographic regions.
5. **Derive actionable policy recommendations** for sustainable housing development, supported by transparent XAI evidence.

4. Literature Review

4.1 Machine Learning in Real Estate Valuation

Machine learning has been extensively applied to real estate price prediction. Park & Bae (2015) demonstrated that ensemble methods outperform traditional hedonic pricing models. Antipov & Pokryshevskaya (2012) applied random forests to mass appraisal, achieving significant accuracy improvements over linear regression. More recently, gradient boosting methods such as XGBoost (Chen & Guestrin, 2016) have shown state-of-the-art performance in tabular data prediction tasks, including real estate valuation.

4.2 Explainable AI (XAI)

The field of XAI has grown rapidly in response to the increasing deployment of complex ML models in high-stakes domains. Arrieta et al. (2020) provide a comprehensive survey of XAI methods, categorizing them into model-specific and model-agnostic approaches. SHAP (Lundberg & Lee, 2017), grounded in Shapley values from cooperative game theory, provides theoretically sound feature attribution. LIME (Ribeiro et al., 2016) offers local interpretability through locally faithful linear approximations. Molnar (2020) provides a practical framework for applying these methods.

4.3 XAI in Housing and Urban Policy

While XAI has been applied in healthcare (Holzinger et al., 2019), finance (Bussmann et al., 2021), and criminal justice, its application in housing policy remains limited. Kok et al. (2017) highlighted the potential of big data and ML in real estate but noted the need for interpretability. Baldominos et al. (2018) applied ML to urban planning but did not incorporate XAI methods. This research fills the gap by combining state-of-the-art XAI with sustainability-focused housing analysis.

4.4 Sustainable Housing Metrics

Sustainable housing encompasses environmental efficiency, affordability, and equitable access (Choguill, 2007). Key metrics include land use efficiency (the ratio of built-up area to lot size), housing density, and affordability indices (Mulliner et al., 2016). Integrating these metrics into ML-based analysis can provide a more holistic view of housing sustainability.

5. Proposed Methodology

5.1 Dataset

This research will use the **USA Real Estate Dataset** from Kaggle (Sakib, 2023), which contains over 2.2 million real estate listings across all U.S. states. The dataset includes the following features:

Table 1: Dataset Features

Feature	Type	Description
price	float	Listing price in USD
bed	float	Number of bedrooms
bath	float	Number of bathrooms
acre_lot	float	Lot size in acres
house_size	float	House size in square feet
state	string	U.S. state
zip_code	float	ZIP code
city	string	City name
status	string	Listing status

5.2 Data Preprocessing

The preprocessing pipeline will include:

- Removal of non-predictive columns (broker ID, street address, previous sale date)
- Handling missing values (dropping rows with missing critical features)
- Price filtering ($10,000 – $5,000,000) to remove erroneous entries
- Property size bounds (100–20,000 sqft, 1–10 bedrooms/bathrooms)
- Outlier removal using percentile-based filtering (1st–99th percentile)

5.3 Feature Engineering

Sustainability-focused features will be engineered:

- **Land Use Efficiency** = house_size / (acre_lot $\times$ 43,560), clipped to [0, 1]
- **Room Density** = (bed + bath) / (house_size / 1000)
- **Total Rooms** = bed + bath
- **Price per Square Foot** = price / house_size
- **Log Price** = $\log(1 + \text{price})$ for target normalization
- **State Affordability Index** = Z-score of state median price

5.4 Model Development

Four models will be trained and compared:

1. **Linear Regression** — baseline linear model
2. **Decision Tree Regressor** — single interpretable tree model
3. **Random Forest Regressor** — ensemble of decision trees using bagging
4. **XGBoost Regressor** — gradient-boosted ensemble with regularization

All models will predict $\log(1 + \text{price})$ to handle the skewed price distribution.

5.5 Evaluation Framework

- **Standard Metrics**: R^2, MAE, RMSE on held-out test set (80/20 split)
- **5-Fold Cross-Validation**: To assess model stability and generalizability
- **Ablation Study**: Compare performance with and without sustainability-engineered features to quantify their contribution
- **Fairness Analysis**: Evaluate prediction equity across price quartiles and U.S. states

5.6 Explainability Analysis

- **SHAP Analysis**: Global feature importance (beeswarm and bar plots), feature dependence plots with interaction effects, and individual prediction waterfall plots
- **LIME Analysis**: Local explanations for representative instances across the price spectrum
- **Cross-Method Validation**: Compare feature rankings from SHAP, LIME, and built-in feature importance using Spearman rank correlation

5.7 Policy Analysis

- State-level sustainability scoring based on land use efficiency, room density, and affordability
- Regional equity assessment through geographic prediction analysis
- Data-driven policy recommendations backed by XAI evidence

6. Expected Outcomes

1. A **comparative analysis** of four ML models for housing price prediction, demonstrating the trade-off between model complexity and performance.
2. **Transparent feature importance rankings** from multiple XAI methods, identifying the key drivers of housing prices (e.g., location, property size, land use efficiency).
3. An **ablation study** quantifying the predictive value of sustainability-engineered features.
4. A **fairness assessment** revealing potential biases in model predictions across price ranges and geographic regions.
5. **Actionable policy recommendations** for sustainable housing development, including:
   - Promotion of land-efficient development patterns
   - Identification of optimal property configurations for affordability
   - Regional equity insights for equitable policy formulation
   - Transparency frameworks for AI-assisted housing assessment
6. A **state-level sustainability scorecard** ranking U.S. states by composite sustainability metrics.

7. Tools and Technologies

Table 2: Tools and Technologies

Category	**Tools**
Programming Language	Python 3.10+
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, XGBoost
Explainability	SHAP, LIME
Visualization	Matplotlib, Seaborn, Plotly
Statistical Analysis	SciPy
Development Environment	Jupyter Notebook
Version Control	Git, GitHub

8. Proposed Timeline

Table 3: Project Timeline

Week	Phase	Activities
1–2	Data Collection & Preprocessing	Dataset acquisition, cleaning, missing value handling, outlier removal
3–4	Feature Engineering	Develop sustainability metrics, exploratory data analysis, correlation analysis
5–6	Model Development	Train and tune four ML models, hyperparameter optimization
7–8	Model Evaluation	Cross-validation, ablation study, fairness analysis
9–10	XAI Analysis	SHAP analysis, LIME explanations, cross-method validation
11–12	Policy Analysis & Report	Sustainability scoring, policy recommendations, final report writing

9. Significance of the Study

This research contributes to the growing body of work at the intersection of AI, sustainability, and public policy:

- **Methodological contribution**: Demonstrates a systematic framework for applying XAI to large-scale housing data, combining multiple explainability methods with rigorous evaluation.
- **Practical contribution**: Provides policymakers with transparent, interpretable insights for housing policy decisions, moving beyond opaque ML predictions.
- **Sustainability contribution**: Introduces sustainability-focused feature engineering for real estate analysis, quantifying the value of metrics like land use efficiency and room density.
- **Fairness contribution**: Evaluates prediction equity across geographic and socioeconomic dimensions, an essential requirement for policy deployment.

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